5 error control coding

Problem 5.1 : consider the following code with 4 codewords of length 6

$$C = \{ 000000,\ 001110,\ 110100,\ 111010 \}$$

→ difference, not dmin but used for it

a) hamming distance: 3, 3, 4, 4, 3, 3

b) d_{min} = $d_{min}(3, 3, 4, 4, 3, 3) = 3$

c) maximum # of errors this code can detect : $e_{detect} = d - 1 = 3 - 1 = 2$ ← $d_{min} \geq e_d + 1$

$d_{min} - 1 \geq e_d$

d) maximum # of errors this code can correct : $e_{correct} = \lfloor \frac{d-1}{2} \rfloor = 1$ ← $d_{min} \geq 2e_c + 1$

↳ round down for max so dont overshoot

$d_{min} - 1 \geq 2e_c$

$\frac{d_{min} - 1}{2} \geq e_c$

e) can it simultaneously detect 2 errors & correct 1 ? justify w dmin

$e_c + e_d \leq d_{min} - 1$

$1 + 2 \leq 3 - 1$

$3 \leq 2$ ✗

no you cannot)

$e_c \leq e_d$

$1 \leq 2$ ✓

Problem 5.2 : a network link has a raw BER of $p = 10^{-3}$ (1 bit per 100,000 bits) . frames carry $k = 1000$ data bits

a) what is the probability that a 1000 bit frame contains at least 1 bit error

frame error probability :

$P_{frame} \approx 1 - (1-P)^n \approx np$ → $np = 1000 \cdot 10^{-3} = 0.01$

P = probability of a bit flipping → $P = 10^{-3}$

n = bits in a frame → $n = 1000$

b) (detection + ARQ) a single parity bit is added ($n = 1001$) what is code rate? undetected error occurs when 2+ errors in a frame, estimate probability of undetected error

code rate: $R = \frac{\text{data bits}}{\text{all bits}} = \frac{1000}{1001} = 0.999$

Probability of undetected error $= \binom{1001}{2} p^2 (1-P)^{999}$

↓

5×10^{-5}

c) (error correction) a hamming code uses r check bits : $n = k + r = 1000 + 10 = 1010$. what is code rate? compare to parity code.

R = code rate $= \frac{1000}{1010} = 0.990$

d) which scheme provides better throughput for this channel, why?

throughput = code rate × probability of success

↳ Parity + ARQ = $0.999 \times 0.99 = 0.989$

Problem 5.3 : message $M = 1101$ and the generator polynomial is $G(x) = x^3 + x + 1$

1011

a) write out the dividend $x^3 M(x)$ (append 3 0s to M)

← times to move everything to left 3

1101 → 1101 000

↖ dividend

b) Perform the modulo 2 long division of $x^3 M(x)$ by $G(x)$ to find remainder $R(x)$

$R(x) = \frac{x^3 M(x)}{G(x)} = \frac{1101000}{1011}$ →

1101
1011
0110

c) write out the transmitted frame T

M(x) R(x)

Original message + append remainder = 1101 001

1101 + 001

d) Suppose the received frame is T with bit 3 flipped. show that the remainder after dividing by G(x) is non zero